Time Series Analysis

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Outline of the lecture

- Regression based methods, 1st part:
  - Introduction (Sec. 3.1)
  - The General Linear Model, including OLS-, WLS-, and ML-estimates (Sec. 3.2)
  - Prediction in the General Linear Model (Sec. 3.3)
  - Examples...

General form of the regression model

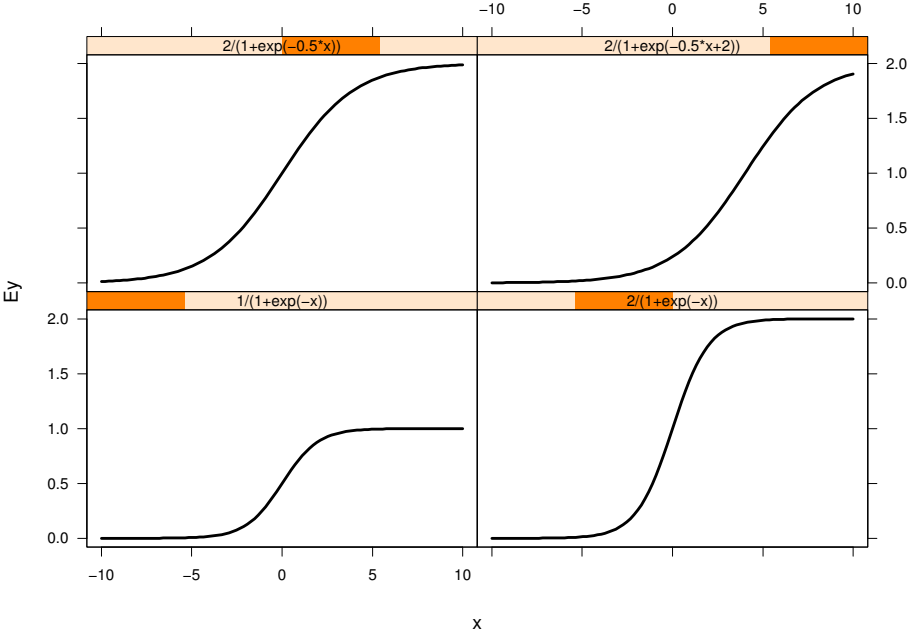
$$Y_t = f(\boldsymbol{X}_t, t; \boldsymbol{\theta}) + \varepsilon_t$$

Where:

- Y_t is the output we aim to model
- $\boldsymbol{X}_t$ indicates the p independent variables $\boldsymbol{X}_t = (X_{1t}, \cdots, X_{pt})^T$
- t is the time index
- $\boldsymbol{\theta}$ indicates m unknown parameters $(\theta_1, \cdots, \theta_m)^T$
- ε_t is a sequence of random variables with mean zero, variance σ_t^2, and $\mathrm{Cov}[\varepsilon_{t_i}, \varepsilon_{t_j}] = \sigma_t^2 \Sigma_{ij}$

For now we restrict the discussion to the case where $\boldsymbol{X}_t$ is non-random and thus we write $\boldsymbol{x}_t$ instead of $\boldsymbol{X}_t$.

$$Y_t = \theta_1/(1 + \exp(-\theta_2 x_t + \theta_3)) + \varepsilon_t$$



Ordinary least squares (OLS) estimates

Observations:

$$(y_1, \boldsymbol{x}_1), (y_2, \boldsymbol{x}_2), \cdots, (y_n, \boldsymbol{x}_n)$$

Ordinary Least Square (unweighted) estimates are found from

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} S(\boldsymbol{\theta})$$

where

$$S(\boldsymbol{\theta}) = \sum_{t=1}^{n} [y_t - f(\boldsymbol{x}_t; \boldsymbol{\theta})]^2 = \sum_{t=1}^{n} \varepsilon_t^2(\boldsymbol{\theta})$$

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For the unweighted method to result in reliable estimates, the errors must be assumed to all have the same variance and be mutually uncorrelated.

OLS – Variance of error and estimates

If the model errors ε_t are i.i.d.

- The variance of the model errors is estimated as:

$$\widehat{\sigma}^2 = \frac{S(\widehat{\boldsymbol{\theta}})}{n-p}$$

  where p is the number of estimated parameters.

- The variance-covariance matrix of the estimates is approximately

$$V[\widehat{\boldsymbol{\theta}}] = 2\widehat{\sigma}^2 \left[\frac{\partial^2}{\partial^2 \boldsymbol{\theta}} S(\boldsymbol{\theta}) \right]^{-1} \Bigg|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}}$$

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Yes, as it can be written as

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It is linearity in the parameters that matters!

General Linear Model - sub classes

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Comments

- For ANOVA and ANCOVA the treatments must be coded into a number of x-variables.
- Some examples in the book

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- For the general linear model a closed-form solution exists.
  For all observations the model equations are written as:

$$\begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{x}_1^T \\ \vdots \\ \boldsymbol{x}_n^T \end{bmatrix} \boldsymbol{\theta} + \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix} \quad or \quad \boldsymbol{Y} = \boldsymbol{x}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$$

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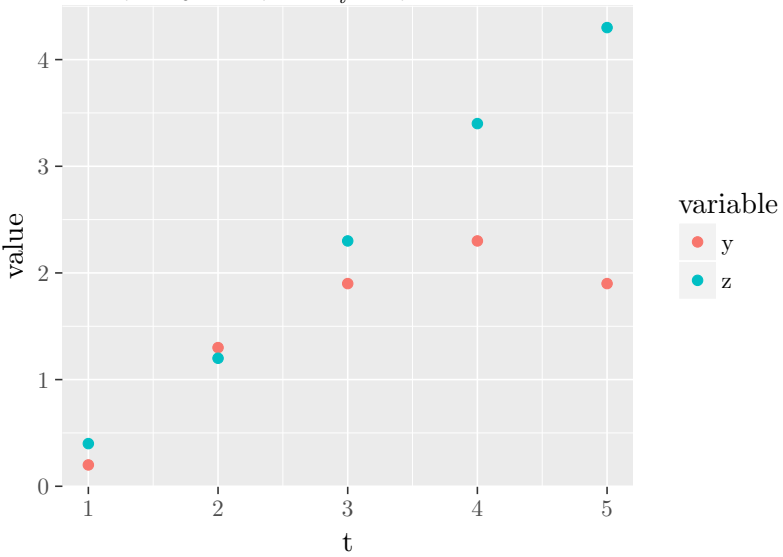
  i.e. we want to minimize $S(\boldsymbol{\theta}) = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}$
- The solution is $\boxed{\widehat{\boldsymbol{\theta}} = (\boldsymbol{x}^T \boldsymbol{x})^{-1} \boldsymbol{x}^T \boldsymbol{Y}}$ (if $\boldsymbol{x}$ has full rank)
- $\widehat{\sigma}^2 = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}/(n-p)$ and $V[\widehat{\boldsymbol{\theta}}] = \widehat{\sigma}^2 (\boldsymbol{x}^T \boldsymbol{x})^{-1}$

Example

Data:

t	y	z
1	0.2	0.4
2	1.3	1.2
3	1.9	2.3
4	2.3	3.4
5	1.9	4.3

Model: $Y_t = \theta_0 + \theta_1 z_t + \theta_2 z_t^2 + \varepsilon_t$



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$$\begin{bmatrix} 0.2 \\ 1.3 \\ 1.9 \\ 2.3 \\ 1.9 \end{bmatrix} = \begin{bmatrix} 1 & 0.4 & 0.16 \\ 1 & 1.2 & 1.44 \\ 1 & 2.3 & 5.29 \\ 1 & 3.4 & 11.56 \\ 1 & 4.3 & 18.49 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \end{bmatrix}$$

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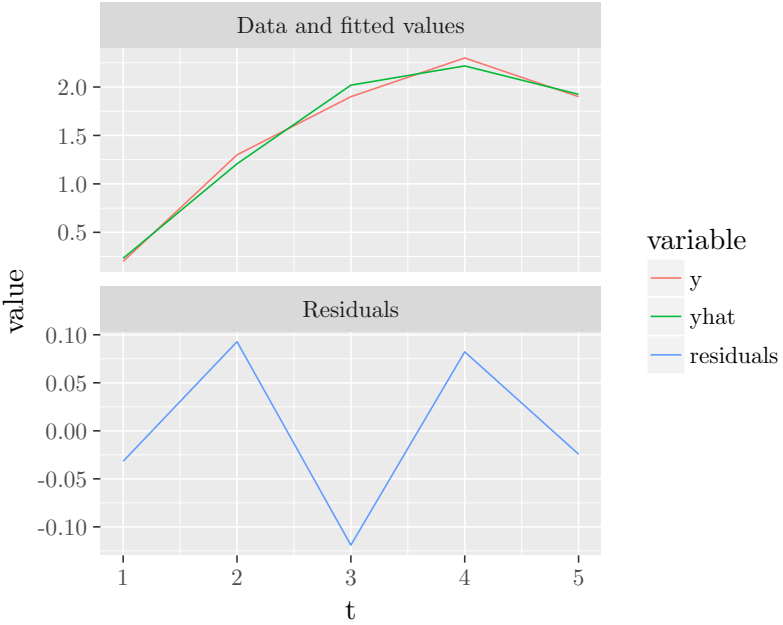
$$\widehat{\boldsymbol{\theta}} = (\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{Y} = \begin{bmatrix} -0.38 \\ 1.62 \\ -0.25 \end{bmatrix}$$

Plot of the model fit



Properties of the OLS-estimator of a GLM

- It is a linear function of the observations $\boldsymbol{Y}$ (and $\widehat{\boldsymbol{Y}}$ is thus a linear function of the observations)
- It is unbiased, i.e. $E[\widehat{\boldsymbol{\theta}}] = \boldsymbol{\theta}$
- $V[\widehat{\boldsymbol{\theta}}] = E\left[(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T\right] = \sigma^2(\boldsymbol{x}^T\boldsymbol{x})^{-1}$
- $\hat{\theta}$ is BLUE (Best Linear Unbiased Estimator), which means that it has the smallest variance among all estimators which are a linear function of the observations.

WLS-estimates

- Equation for all observations: $\boldsymbol{Y} = \boldsymbol{x\theta} + \boldsymbol{\varepsilon}$
- $E[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $V[\boldsymbol{\varepsilon}] = E[\boldsymbol{\varepsilon\varepsilon}^T] = \sigma^2\boldsymbol{\Sigma}$, where $\boldsymbol{\Sigma}$ is known
- We want to minimize $(\boldsymbol{Y} - \boldsymbol{x\theta})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{Y} - \boldsymbol{x\theta})$

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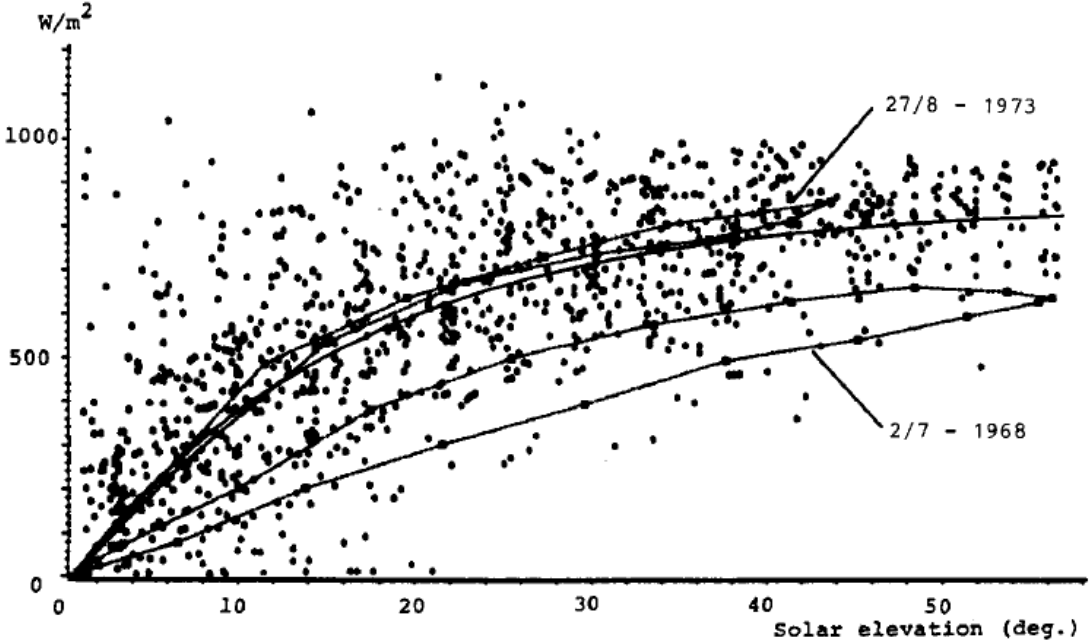
- We want to minimize $(\boldsymbol{Y} - \boldsymbol{x}\boldsymbol{\theta})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{Y} - \boldsymbol{x}\boldsymbol{\theta})$
- The solution is
$$\widehat{\boldsymbol{\theta}} = (\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{Y}$$
(if $\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x}$ is invertible)
- An estimate of σ^2 is
$$\widehat{\sigma}^2 = \frac{1}{n-p}(\boldsymbol{Y} - \boldsymbol{x}\widehat{\boldsymbol{\theta}})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{Y} - \boldsymbol{x}\widehat{\boldsymbol{\theta}})$$

Example WLS/OLS

- H. Madsen & P. Thyregod (1988). *Modelling the Time Correlation in Hourly Observations of Direct Radiation in Clear Skies*. Energy and Buildings, **11**, 201–211.
- See the examples in the book.

Example WLS/OLS: Clear sky radiation



Example WLS/OLS: Clear sky radiation

$$I_N(h(t)) = a_N(1 - \exp(-b_N h(t))) + \varepsilon_N(t) \tag{1}$$

$$\varepsilon_N(t) \sim N(\mathbf{0}, \sigma^2 \mathbf{\Sigma}) \tag{2}$$

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Suggested variance structures

- i.i.d

$$\boldsymbol{\Sigma} = \boldsymbol{I}$$

- Only correlation

$$\Sigma_{ij} = \rho^{|t_i - t_j|}$$

- Only variance

$$\Sigma_{ij} = \frac{1}{\sin(h(t_i))\sin(h(t_j))}$$

- Both correlation and variance

$$\Sigma_{ij} = \frac{\rho^{|t_i - t_j|}}{\sin(h(t_i))\sin(h(t_j))}$$

Maximum Likelihood (ML) - estimates

- We now assume that the observations are Gaussian:

$$\boldsymbol{Y} \sim \mathrm{N}_n(\boldsymbol{x}\boldsymbol{\theta}, \sigma^2\boldsymbol{\Sigma})$$

- $\boldsymbol{\Sigma}$ is assumed known
- The ML-estimator is (here) the same as the WLS-estimator:

$$\widehat{\boldsymbol{\theta}} = (\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{Y}$$

- The ML-estimator for σ^2 is

$$\widehat{\sigma}^2 = \frac{1}{n}(\boldsymbol{Y} - \boldsymbol{x}\widehat{\boldsymbol{\theta}})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{Y} - \boldsymbol{x}\widehat{\boldsymbol{\theta}})$$

Example WLS/OLS: Solar radiation

Structure	$\hat{a}_N$	$\hat{b}_N$	$\hat{\sigma}_N^2$
Identity	798.6 (13.1)	0.0798 (0.0043)	34631.8
Correlation	827.8 (7.3)	0.0551 (0.0022)	25180.1
Variance	822.6 (5.2)	0.0706 (0.0012)	3947.6
Var. and Cor.	842.3 (7.4)	0.0614 (0.0013)	2302.1

Numbers in paranthesis are standard deviations.

Properties of the ML-estimator

- It is a linear function of the observations which now implies that it is normally distributed.
- It is unbiased, i.e. $E[\widehat{\boldsymbol{\theta}}] = \boldsymbol{\theta}$ and
- The variance $V[\widehat{\boldsymbol{\theta}}] = E[(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T] = (\boldsymbol{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{x})^{-1} \sigma^2$;
- It is an efficient estimator (minimum variance of unbiased estimators).

Unknown $\boldsymbol{\Sigma}$

Relaxation algorithm:

a) Select a value for $\boldsymbol{\Sigma}$ (e.g. $\boldsymbol{\Sigma} = \boldsymbol{I}$).

b) Find the estimates for this value of $\boldsymbol{\Sigma}$ e.g. by solving the normal equations.

c) Consider the residuals $\{\widehat{\boldsymbol{\varepsilon}}_t\}$ and calculate the correlation and variance structure of the residuals. Then select a new value for $\boldsymbol{\Sigma}$ which reflects that correlation and variance structure.

d) Stop if convergence - otherwise go to b).

See (Goodwin and Payne, 1977) for details.

Prediction

Theorem 3.8 and 3.9

- If the expected value of the squared prediction error is to be minimized, then
- the expected mean $E[Y|\boldsymbol{X} = \boldsymbol{x}]$ is the optimal predictor.

Prediction in the general linear model

- Known parameters:

$$\widehat{Y}_{t+\ell} = E_\theta[Y_{t+\ell}|\boldsymbol{X}_{t+\ell} = \boldsymbol{x}_{t+\ell}] = \boldsymbol{x}_{t+\ell}^T\boldsymbol{\theta}$$

$$V_\theta[Y_{t+\ell} - \widehat{Y}_{t+\ell}] = V_\theta[\varepsilon_{t+\ell}] = \sigma^2$$

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- Estimated parameters:

$$\widehat{Y}_{t+\ell} = E_{\widehat{\theta}}[Y_{t+\ell}|\boldsymbol{X}_{t+\ell} = \boldsymbol{x}_{t+\ell}] = \boldsymbol{x}_{t+\ell}^T\widehat{\boldsymbol{\theta}}$$

$$V_{\widehat{\theta}}[Y_{t+\ell} - \widehat{Y}_{t+\ell}] = V_{\widehat{\theta}}[\varepsilon_{t+\ell} + \boldsymbol{x}_{t+\ell}^T(\theta - \widehat{\theta})] = \widehat{\sigma}^2[1 + \boldsymbol{x}_{t+\ell}^T(\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}_{t+\ell}]$$

Prediction in the general linear model – continued

- In practice we have to use an estimate of σ and therefore a $100(1-\alpha)\%$ prediction interval of a future value is calculated as:

$$\widehat{Y}_{t+\ell} \pm t_{\alpha/2}(n-p)\hat{\sigma}\sqrt{1+\boldsymbol{x}_{t+\ell}^T(\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}_{t+\ell}}$$

  where $t_{\alpha/2}(n-p)$ refers to the $\alpha/2$'th quantile of the t-distribution with $n-p$ degrees of freedom.
- For $n-p$ large, percentiles from the normal distribution can be used.

Highlights

- When ε_t are i.i.d then the variance of the model errors is estimated as:

$$\widehat{\sigma}^2 = \frac{S(\widehat{\boldsymbol{\theta}})}{n-p}$$

- OLS estimator:

$$\widehat{\boldsymbol{\theta}} = (\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{Y}$$

- WLS estimator (When $V[\boldsymbol{\varepsilon}] = E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T] = \sigma^2\boldsymbol{\Sigma}$)

$$\widehat{\boldsymbol{\theta}} = (\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{Y}$$

- ML estimator for $\widehat{\theta}$ is the same as the WLS estimator.
- but not for $\widehat{\sigma}^2$
- The expected mean $E[Y|\boldsymbol{X} = \boldsymbol{x}]$ is the optimal predictor.
- Prediction in GLM

$$\widehat{Y}_{t+\ell} \pm t_{\alpha/2}(n-p)\hat{\sigma}\sqrt{1 + \boldsymbol{x}_{t+\ell}^T(\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}_{t+\ell}}$$